FACULTY OF COMPUTING
UTM Johor Bahru

Semester II 2025/2026

Subject	:	Special Topic in Data Engineering (SECP3843)
Task	:	Tutorial 4: Claude & Pipeline
Due	:	23.6.2026
Submission	:	Present and demo
Assignment Type	:	Group (3 students)
Technology	:	Claude

1. Select a tutorial in website:

   Tutorial 1

   https://www.youtube.com/watch?v=e8MTnEBdEz4 **(From Claude to Pipelines: AI-Assisted Data Engineering)**

   Tutorial 2

   https://www.youtube.com/watch?v=GDmEgrX_ZQc (The AI Workflow for Data Engineering)

   Tutorial 3

   https://www.youtube.com/watch?v=Pl3aaM2XHpo **(The End of Rule-Based Data Quality? How to Use AI Agents Instead)**

   Tutorial 4

   https://www.youtube.com/watch?v=QQCqQTvtz8w (AI for Data Pipelines in HighByte Intelligence Hub)

   Tutorial 5

   https://www.youtube.com/watch?v=qGu7hDaJG1A **(ETL data pipeline with an AI Agent)**

   i. Writing Report content

      a. What is AI-assisted data engineering
      b. What is Claude/Ai agent use in the tutorial?
      c. Steps to do prompt and coding
      d. Reflection

Rubric for Data Engineering Tutorial

Assessment Components

Component	**Weight**
Written Report	50%

Component	**Weight**
Solution Demo	40%
Reflection	10%
Total	**100%**

📝 1. Written Report (50%)

Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)
Understanding of AI-Assisted Data Engineering	Comprehensive explanation with clear concepts, examples, and relevance to modern data engineering	Clear explanation with minor gaps	Basic explanation with limited clarity	Incorrect or unclear explanation
Understanding of Claude	Clear, detailed explanation of Claude's role in data engineering (automation, prompting, pipelines)	Good explanation but lacks depth	Basic description only	Incorrect or unclear understanding
Steps: Prompt and Code Working with Data	Clearly explains end-to-end pipeline (data ingestion, transformation, output) with code justification	Steps are clear but limited explanation	Basic steps shown but not well explained	Incomplete or incorrect steps
Integration of AI in Pipeline	Demonstrates strong understanding of how AI supports data tasks (e.g., automation, transformation, debugging)	Shows integration but limited explanation	Minimal integration shown	No clear AI integration
Results & Explanation	Results are clearly presented, interpreted, and linked to objectives	Results presented with limited explanation	Basic results only	No meaningful results
Report Organization & Academic Writing	Well-structured, clear flow, academic tone, proper formatting	Generally well-organized	Acceptable but lacks clarity	Poorly structured and unclear

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💻 2. Solution Demo (40%)

Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)
Implementation Accuracy	Fully working pipeline with correct logic and execution	Minor errors but functional	Partially working	Not working
Use of AI Tool (Claude)	Effectively uses Claude for generating, improving, or debugging code	Uses Claude but not optimally	Minimal use of AI	No use of AI tool

Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)
Code Quality	Clean, well-structured, commented, reusable	Mostly clean with minor issues	Basic structure	Poor or messy code
Data Handling	Proper data ingestion, cleaning, and transformation demonstrated clearly	Mostly correct handling	Limited processing	Incorrect or missing handling
Pipeline Design	Clear pipeline structure (input → processing → output)	Pipeline exists but not well explained	Basic pipeline	No clear pipeline
Demo Presentation	Clear, confident, and professional explanation	Clear but less engaging	Basic explanation	Poor or unclear demo

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🔍 3. Reflection (10%)

Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)
Learning Reflection	Deep insight into learning AI-assisted workflows and data engineering concepts	Good reflection with some insights	Basic reflection	Very limited reflection
Use of AI Insight	Clearly reflects on benefits/limitations of using Claude in development	Some discussion of AI use	Minimal mention	No reflection on AI use
Future Improvement	Strong, realistic ideas for improving pipeline and AI usage	Some suggestions	Basic ideas	No suggestions